

Reef site and habitat influence effectiveness of *Acropora Palmata* restoration and its microbiome in the Florida Keys

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Supplementary Table 1. The number of *Acropora palmata* samples used for 16S rRNA microbiome data analysis by genet, reef, and date of collections.

<u>Reef/Month</u>	<u>CN2</u>	<u>CN4</u>	<u>HS1</u>	<u>ML2</u>	<u>Grand Total</u>
Carysfort	32	8	32	9	81
2/28/19	8	4	9	5	26
6/25/19	9	4	9	3	25
9/25/19	8		6		14
11/19/19	7		8	1	16
North Dry Rocks	35	32	31	34	132
3/4/19	9	9	8	9	35
6/26/19	9	9	9	9	36
9/25/19	9	6	6	8	29
11/19/19	8	8	8	8	32
Pickles Reef	35	35	34	29	133
3/1/19	9	9	8	9	35
7/2/19	9	9	9	9	36
9/26/19	9	9	8	5	31
11/18/19	8	8	9	6	31
Grand Total	102	75	97	72	346

Supplementary Table 2. Results from the Cox proportional hazards regression model to examine the association between the survival of genets and across reef. For genet comparisons are relative to CN2 and for reef (North Dry Rocks [NDR], and Pickles Reef [PR]) relative to Carysfort.

Factor(s)	Coef	Hazard ratio= exp(coef)	se(coef)	z statistic	p-value
Genet CN4	1.7115	5.5373	0.2703	6.331	2.43e-10
Genet HS1	0.1276	1.1361	0.3097	0.412	0.68
Genet ML2	1.3234	3.7563	0.3094	4.277	1.89e-05
Reef NDR	-2.1169	0.1204	0.2359	-8.9739	< 2e-16
Reef PR	-2.5523	0.0779	0.2706	-9.431	< 2e-16

Supplementary Table 3. *Acropora palmata* surveys conducted from 2018 - 2019. The surveys were analyzed in three ways: (1) All fragments within a cluster were surveyed (2) All fragments grouped by cluster, and (3) only fragments that were sampled for tissue analysis. Carysfort Reef [CF], North Dry Rocks [NDR], and Pickles Reef [PR].

		All fragments				By Cluster				Fragments for tissue samples			
Reef	Genet	Total N	N of Events	Censored		Total N	N of Events	Censored		Total N	N of Events	Censored	
				N	Survivorship			N	Survivorship			N	Survivorship
CF	CN2	36	14	22	61.10%	20	5	15	75.00%	9	1	8	88.90%
	CN4	31	28	3	9.70%	13	10	3	23.10%	4	3	1	25.00%
	HS1	32	13	19	59.40%	20	4	16	80.00%	12	2	10	83.30%
	ML2	20	18	2	10.00%	13	11	2	15.40%	6	5	1	16.70%
Overall		119	73	46	38.70%	66	30	36	54.50%	31	11	20	64.50%
NDR	CN2	38	6	32	84.20%	20	2	18	90.00%	10	1	9	90.00%
	CN4	42	17	25	59.50%	23	11	12	52.20%	12	7	5	41.70%
	HS1	58	2	56	96.60%	20	0	20	100.00%	9	0	9	100.00%
	ML2	20	1	19	95.00%	20	1	19	95.00%	11	0	11	100.00%
Overall		158	26	132	83.50%	83	14	69	83.10%	42	8	34	81.00%
PR	CN2	42	0	42	100.00%	21	0	21	100.00%	11	0	11	100.00%
	CN4	42	7	35	83.30%	22	6	16	72.70%	9	3	6	66.70%
	HS1	58	7	51	87.90%	21	1	20	95.20%	12	1	11	91.70%
	ML2	21	4	17	81.00%	21	4	17	81.00%	10	1	9	90.00%
Overall		163	18	145	89.00%	85	11	74	87.10%	42	5	37	88.10%
Overall		440	117	323	73.40%	234	55	179	76.50%	115	24	91	79.10%

Supplementary Table 4. Average sea surface temperatures (SST) and chlorophyll-a (Chl-a) concentrations on the day or week coral tissue sampling took place, respectively. The table is divided by month and reef (Carysfort Reef [CF], North Dry Rocks [NDR], and Pickles Reef [PR]).

Month(s)	Reef	Average SST (°C)	Standard deviation SST °C	Average Chl-a mg m-3	Standard deviation Chl-a mg m-3
Feb-Mar	cf	25.96821	0.1514858	0.6960258	0.4539086
Feb-Mar	ndr	26.45431	0.4208808	2.7002575	0.8883858
Feb-Mar	pr	25.99442	0.2483825	0.3999531	0.1630120
Jun-Jul	cf	29.63623	0.5081026	1.7920917	0.3763544
Jun-Jul	ndr	30.03815	0.3614116	3.0906976	1.6064865
Jun-Jul	pr	29.49291	0.7047354	1.5077771	0.9649057
Sept	cf	29.41868	0.4285187	1.1930631	0.0000000
Sept	ndr	29.05134	0.6739111	1.9975726	0.0000000
Sept	pr	29.12768	0.5098662	0.8388900	0.3133472
Nov	cf	27.62378	0.2019326	0.9899462	0.1509655
Nov	ndr	27.19708	0.3856202	3.2423773	1.7397468
Nov	pr	27.43931	0.4212546	0.5188550	0.2415853

Supplementary Table 5. Results from Wilcoxon signed-rank test to test differences between reefs (Carysfort Reef [CF], North Dry Rocks [NDR], and Pickles Reef [PR]) and months across nutrients analyzed in this study. “Ns” denotes not significant, **= padj<0.01, ***-padj<0.005, and **** padj=<0.0005

Nutrients	Group1	Group2	Statistic	pvalue	p.adj	p.adj.signif
Silica	cf	ndr	153	0.091	0.91	ns
Silica	cf	pr	75.5	0.081	0.891	ns
Silica	ndr	pr	51	0.006	0.072	ns
Nitrite	cf	ndr	108.5	0.884	1	ns
Nitrite	cf	pr	144	0.351	1	ns
Nitrite	ndr	pr	150.5	0.234	1	ns
Nitrate	cf	ndr	137	0.319	1	ns
Nitrate	cf	pr	130	0.707	1	ns
Nitrate	ndr	pr	108	0.649	1	ns
Phosphate	cf	ndr	136	0.338	1	ns
Phosphate	cf	pr	116	0.89	1	ns
Phosphate	ndr	pr	91.5	0.268	1	ns
Silica	Feb-Mar	Jun-Jul	0	0.024	0.576	ns
Silica	Feb-Mar	Sept	55	0.01	0.24	ns
Silica	Feb-Mar	Nov	25	0.88	1	ns
Silica	Jun-Jul	Sept	114	0.000218	0.005232	**
Silica	Jun-Jul	Nov	67	0.405	1	ns
Silica	Sept	Nov	48	0.000159	0.003816	***
Nitrite	Feb-Mar	Jun-Jul	4	0.241	1	ns

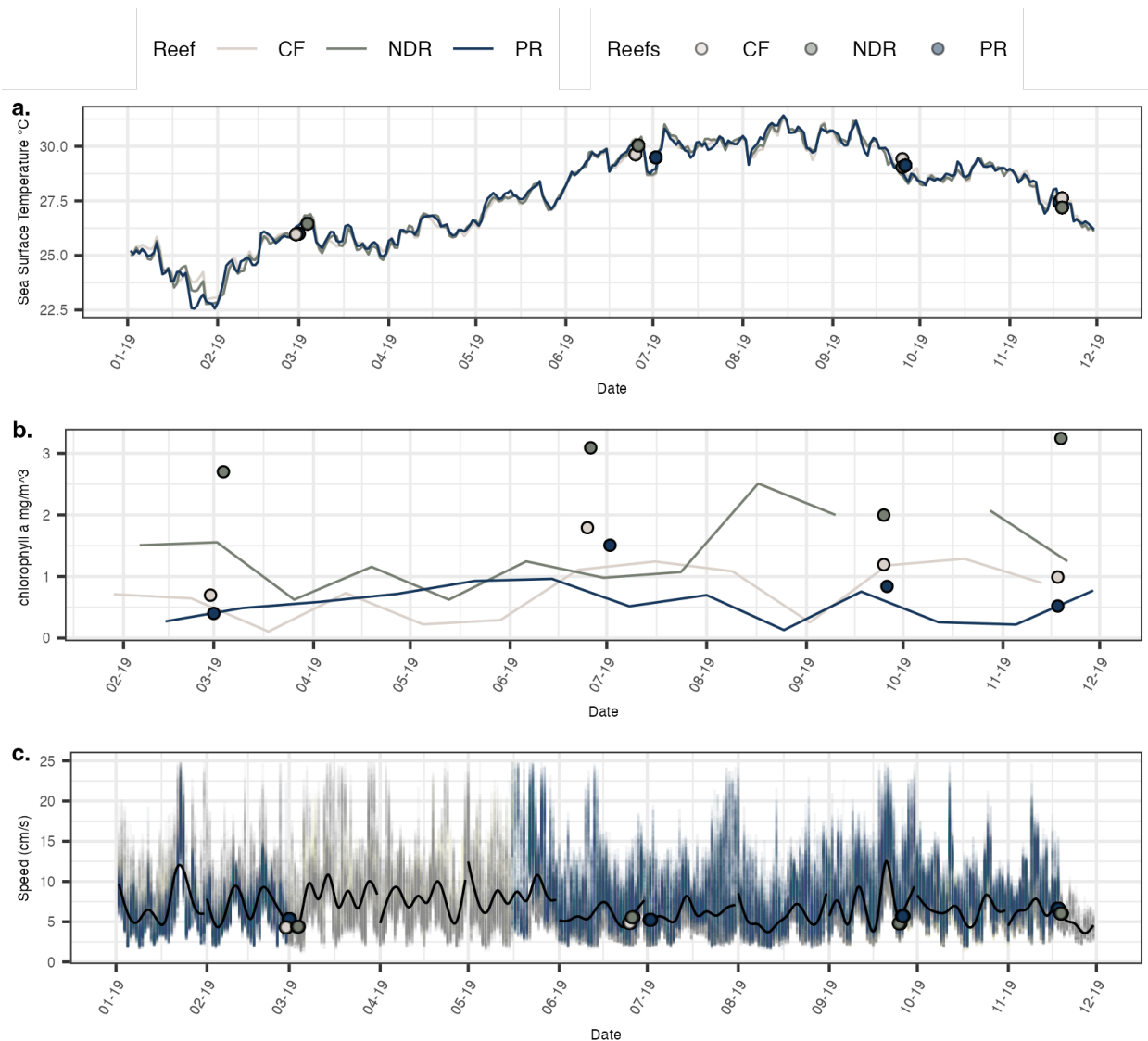
Nitrite	Feb-Mar	Sept	16	0.242	1	ns
Nitrite	Feb-Mar	Nov	36.5	0.364	1	ns
Nitrite	Jun-Jul	Sept	57	1	1	ns
Nitrite	Jun-Jul	Sept	83.5	0.053	1	ns
Nitrite	Sept	Nov	264	0.005	0.12	ns
Nitrate	Feb-Mar	Jun-Jul	3	0.167	1	ns
Nitrate	Feb-Mar	Sept	13	0.151	1	ns
Nitrate	Feb-Mar	Nov	54	0.002	0.048	**
Nitrate	Jun-Jul	Sept	67	0.545	1	ns
Nitrate	Jun-Jul	Nov	108	1.49E-05	0.000358	****
Nitrate	Sept	Nov	336	5.76E-07	1.38E-05	****
Phosphate	Feb-Mar	Jun-Jul	10	0.905	1	ns
Phosphate	Feb-Mar	Sept	27	0.923	1	ns
Phosphate	Feb-Mar	Nov	25	0.88	1	ns
Phosphate	Jun-Jul	Sept	57	1	1	ns
Phosphate	Jun-Jul	Nov	50.5	0.841	1	ns
Phosphate	Sept	Nov	160	0.749	1	ns

Supplementary Table 6. Results from PERMANOVA using the adonis2 vegan function to identify the relationship between microbial composition and the listed factors.

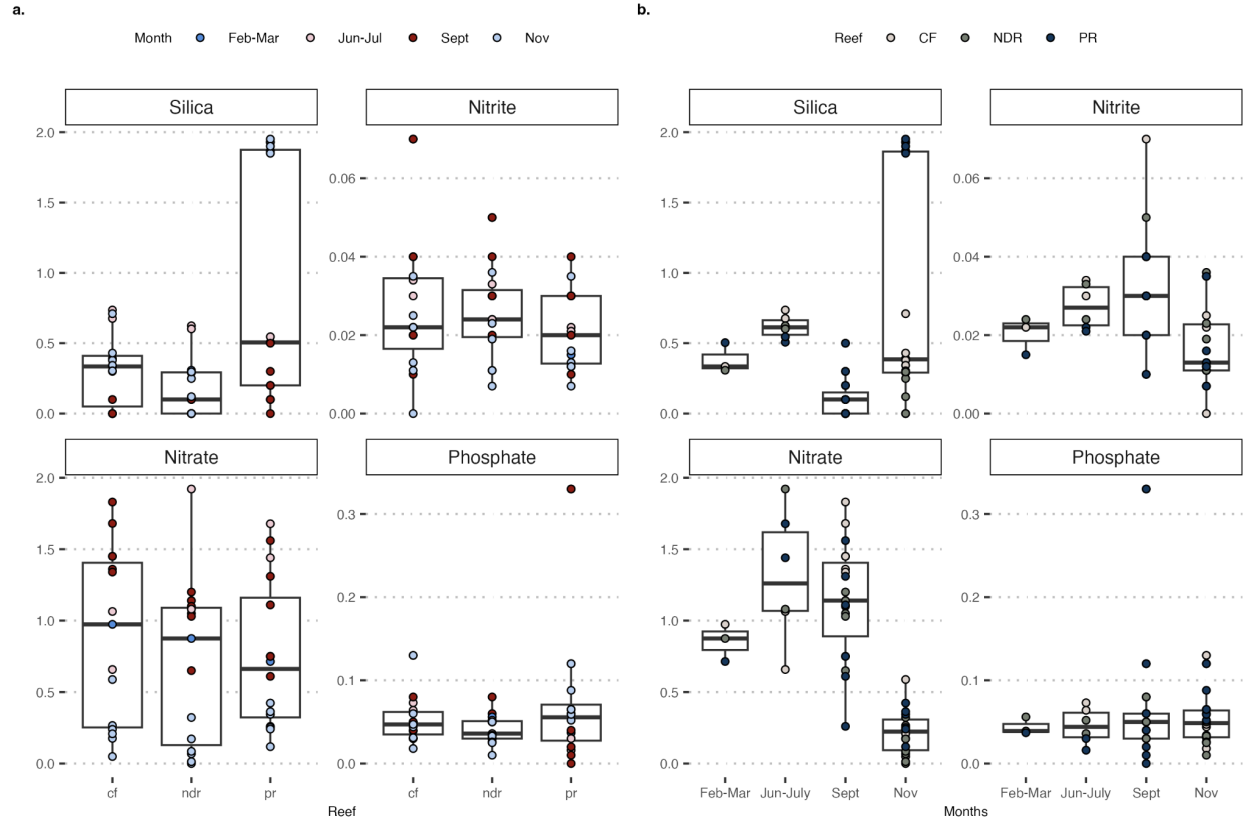
Factor(s)	DF	Sum of Sq	R ²	F	p-value
Month	3	25417.95	0.09	4.296	0.001
Genet	3	11986.15	0.04	2.026	0.001
Reef	2	19285.74	0.06	4.890	0.001
Month:Genet	9	18527.08	0.06	1.044	0.275
Month:Reef	6	22347.75	0.075	1.889	0.001
Genet:Reef	6	14703.86	0.049	1.243	0.009
Month:Genet:Reef	15	32887.72	0.11	1.111	0.036
Residuals	78	153803.14	0.51	NA	NA
Total	122	298959.39	1.00000000	AN	NA

Supplementary Table 7. Results from envifit function on R. The function fits onto an ordination of both continuous variables such as survivorship, and habitat variables as well as factors like genet, reef, and month.

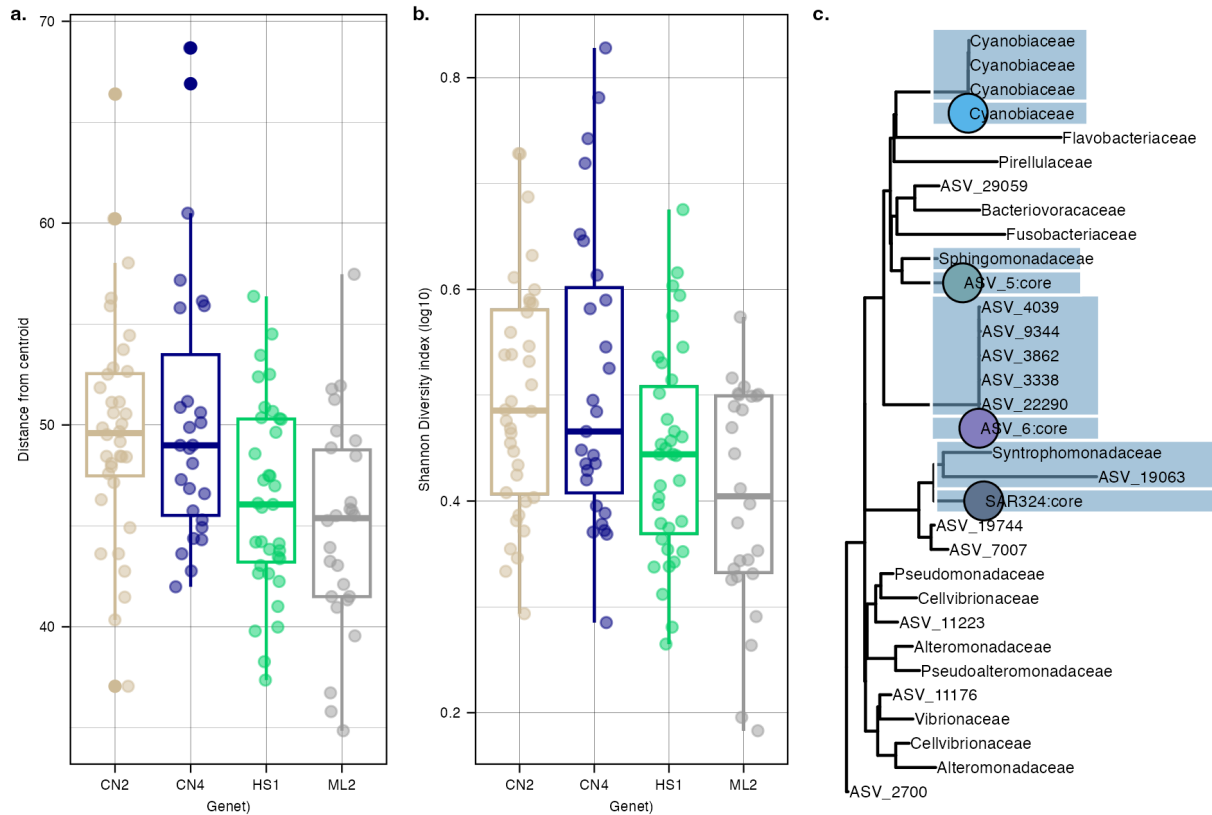
Vector/Factor	R ²	p-value
Survivorship	0.007 1	0.623
Currents	0.0813	0.006
Nitrite	0.4745	0.001
Nitrate	0.2141	0.001
Phosphate	0.0156	0.367
Silica	0.1015	0.001
Sea Surface Temp	0.7158	0.001
Chlorophyll-a	0.4239	0.001
Depth	0.3095	0.001
Genet	0.0082	0.928
Month	0.4055	0.001
Reef	0.3143	0.001



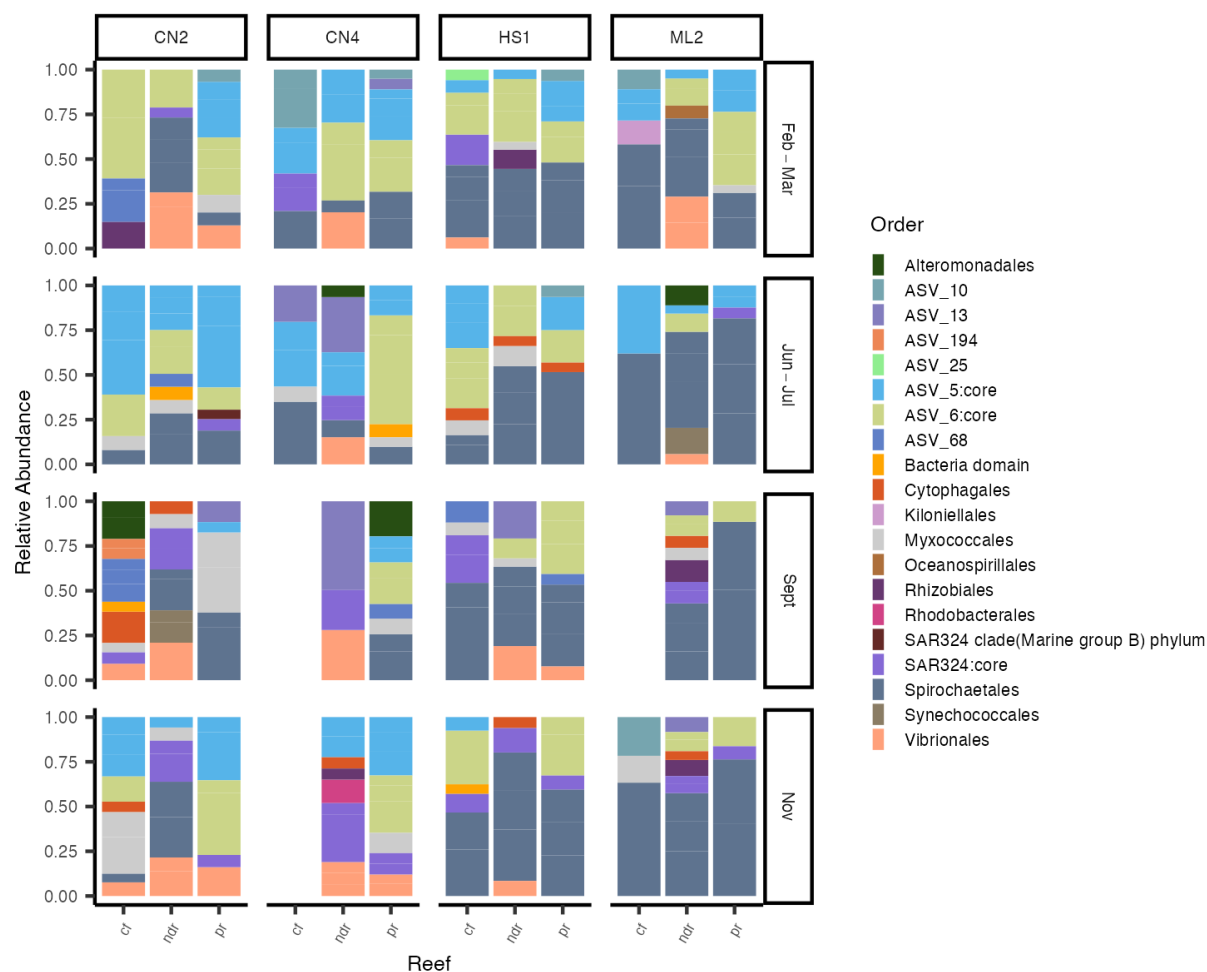
Supplementary Figure 1. Habitat time series in 2019 at three upper Florida Keys sites. (a) sea surface temperature (SST) **(b)** chlorophyll-a and **(c)** current speed collected by tilt current meters. The colors represent the three different reefs and the circles represent field dates when *Acropora palmata* tissue samples were collected.



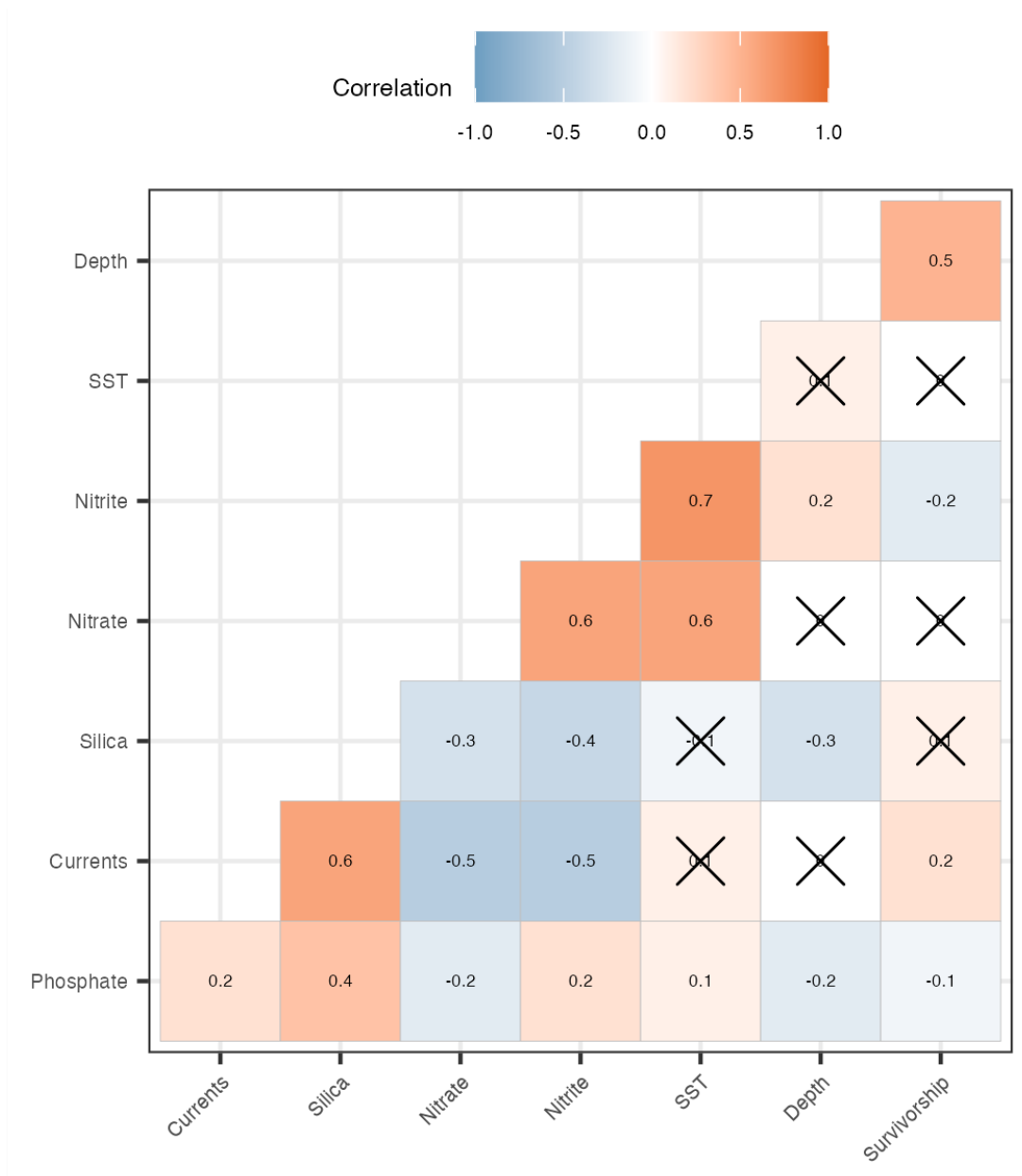
Supplementary Figure 2. Boxplots of nutrient data collections. Water sample collections for nutrients across (a.) coral reefs, and (b.) months, which coincided with *Acropora palmata* tissue sampling. The lower quartile, the median, and the upper quartile correspond to the 25th, 50th, and 75th percentiles, respectively are represented in each boxplot. Outliers are represented as individual points beyond the whiskers which extend to the smallest and largest values within 1.5 times the interquartile range from Q1 and Q3.



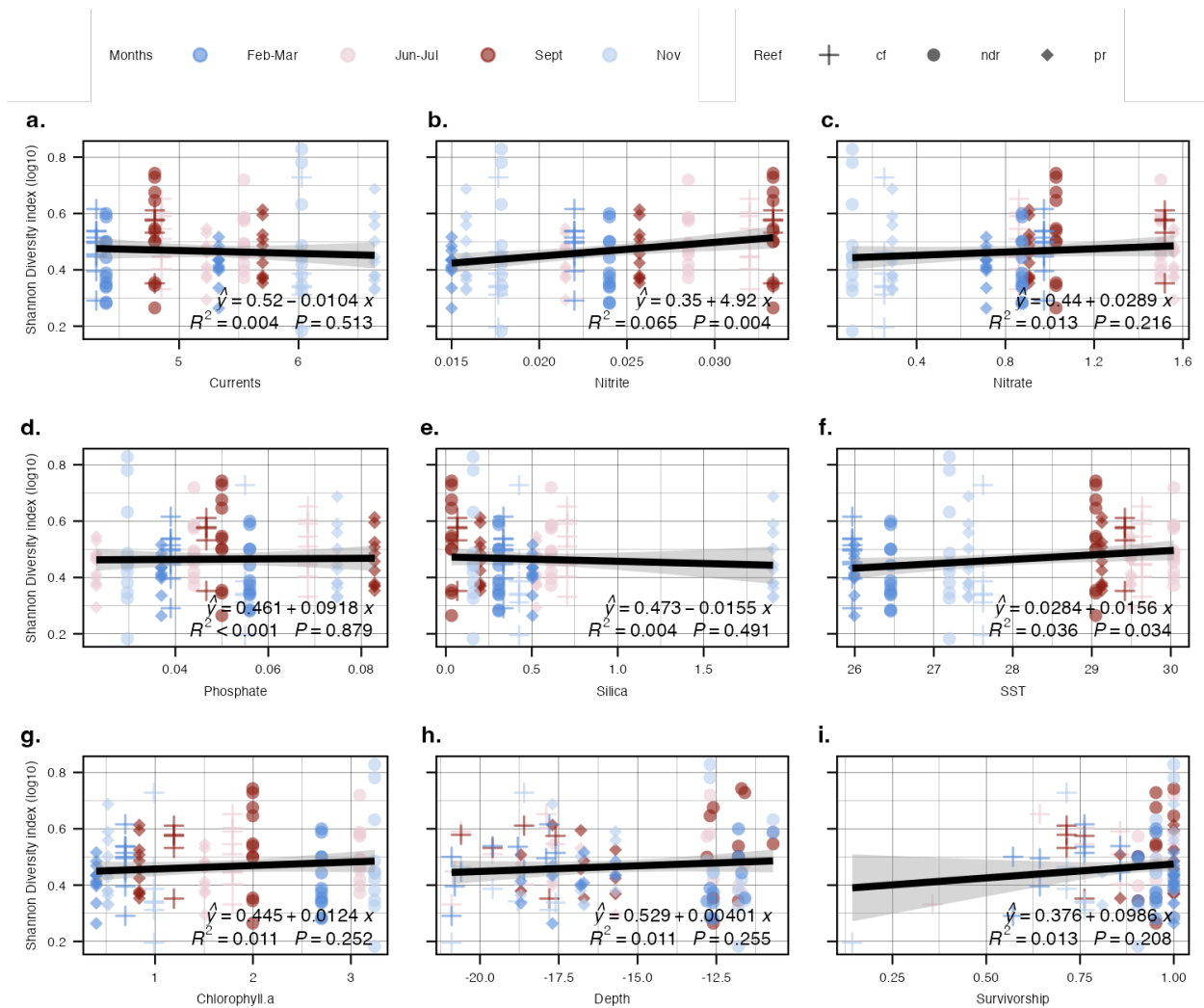
Supplementary Figure 3. Microbial composition of four *Acropora palmata* genets. Differences in (a) Dispersion (i.e., community variability) and (b) Shannon diversity (i.e., alpha-diversity) of the four genets examined. (c) A phylogenetic tree of the four core bacteria members found across the four *A. palmata* genets. Each colored bubble represents a unique ASV. The tree is a subtree constructed from all ASVs and for each core bacteria a sibling and one first cousin, if they existed, were randomly selected for the phylogenetic tree. For figures a-b, the lower quartile, the median, and the upper quartile corresponds to the 25th, 50th, and 75th percentiles, respectively, and are represented in each boxplot. Outliers are represented as individual points beyond the whiskers which extend to the smallest and largest values within 1.5 times the interquartile range from Q1 and Q3.



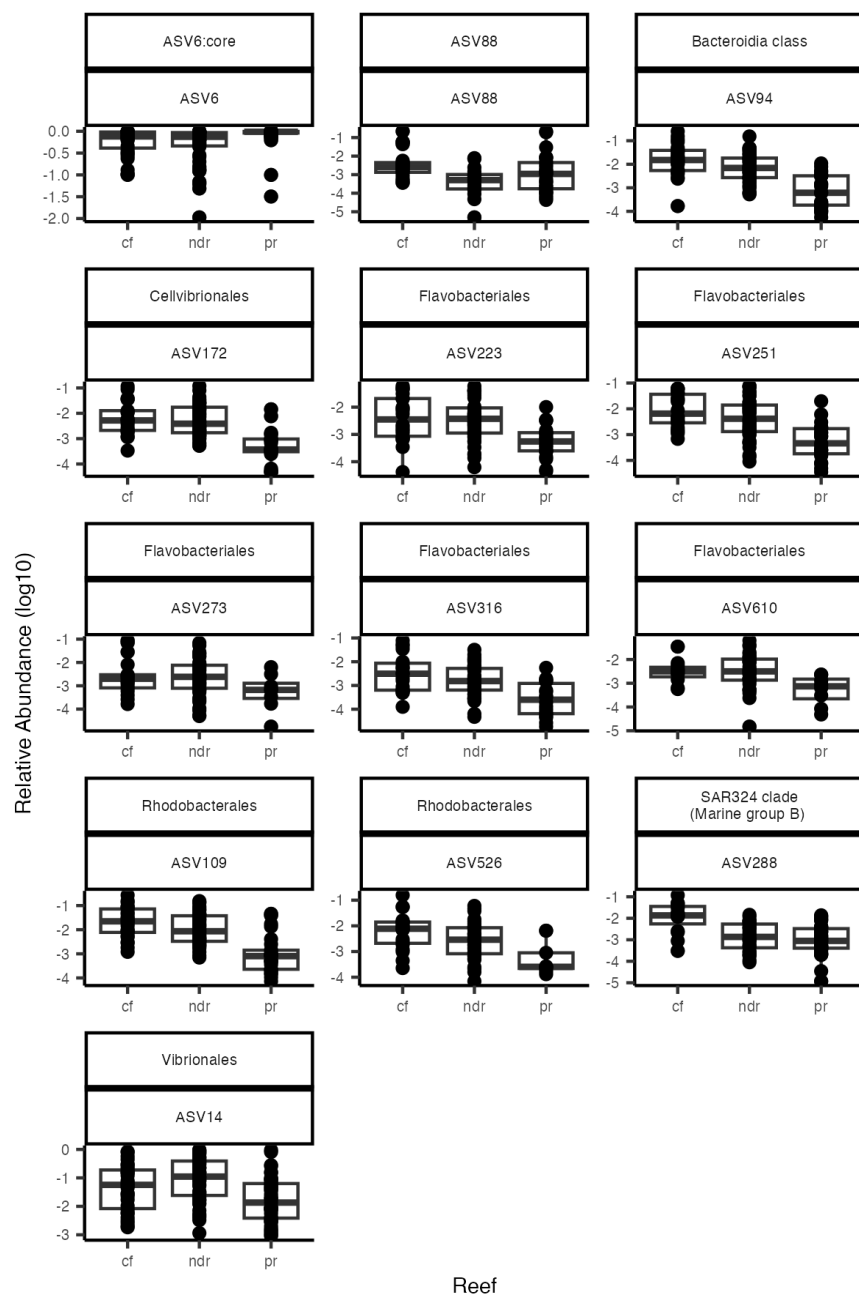
Supplementary Figure 4. Relative abundance of microbial taxa associated with *Acropora palmata*. The data are divided by four genets and three coral reefs (Carysfort Reef [CF], North Dry Rocks [NDR], and Pickles Reef [PR]) across 2019 from February to November. Stacked bar colors indicate different taxonomic orders if the order was not taxonomically classified then the lowest taxonomic assignment was used. Only data with greater than 10% relative abundance were plotted.



Supplementary Figure 5. Correlation matrix between habitat data, and survivorship. Habitat data were correlated and evaluated for significance. The “X” on the figure signifies there was no significance between correlations.

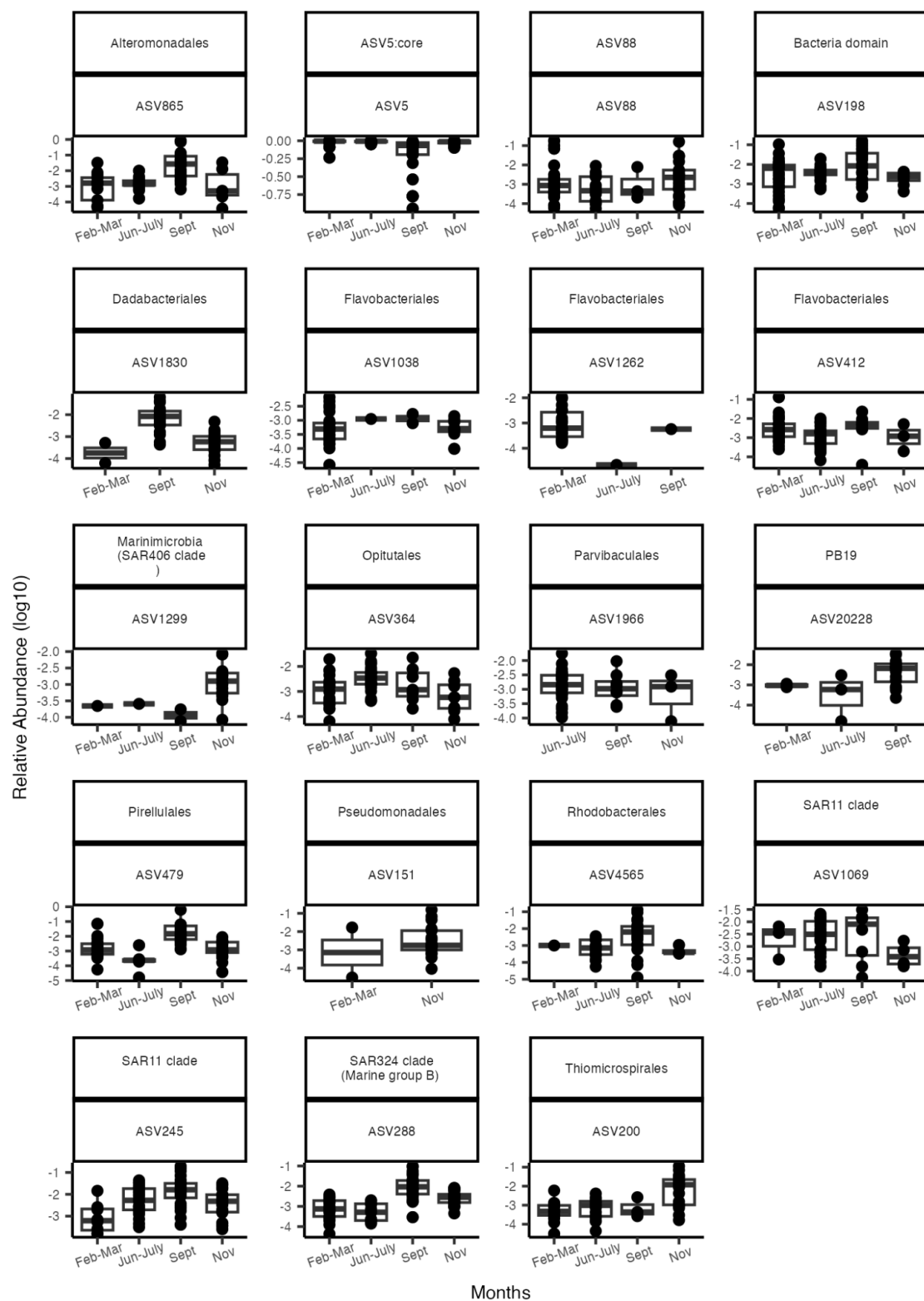


Supplementary Figure 6. *Acropora palmata* microbial Shannon-diversity correlations to habitat factors and survivorship. These included (a) current velocities, (b) nitrite, (c) nitrate, (d) phosphate, (e) silica, (f) sea surface temperature (SST), (g) Chlorophyll a (h) depth, and (i) survivorship. The grey around the lines represents the 95% confidence intervals from the linear model.



Supplementary Figure 7. **Box plots of bacteria taxa assigned the highest importance values by random forest (RF) in the three coral reefs examined (Caysfort Reef [CF], North Dry Rocks [NDR], and Pickles Reef [PR]).** The box plots show relative abundance among the top RF taxa for each coral reef, which were log-transformed for visualization. Each boxplot shows the ASV order or the lowest taxonomic assignment given and ASV number. The lower quartile, the median, and the upper quartile, which correspond to the 25th, 50th, and 75th percentiles, respectively are represented in each boxplot. Outliers are represented as individual points

beyond the whiskers which extend to the smallest and largest values within 1.5 times the interquartile range from Q1 and Q3.



Supplementary Figure 8. **Box plots of bacteria taxa assigned the highest importance values by random forest (RF) during four sampling timepoints.** The box plots show the relative abundance among the top RF taxa and were log-transformed for visualization. Each boxplot

shows the ASV order or the lowest taxonomic assignment given and ASV number. The lower quartile, the median, and the upper quartile, which correspond to the 25th, 50th, and 75th percentiles, respectively are represented in each boxplot. Outliers are represented as individual points beyond the whiskers which extend to the smallest and largest values within 1.5 times the interquartile range from Q1 and Q3.